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# Simulation of New Growth Patterns in Forest Ecosystems with Mycorrhizal Networks

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## 1. Introduction

**Mycorrhizal Networks (MNs)** are ubiquitous underground systems of fungal species linking plant roots, serving as critical avenues for the transfer of carbon, nutrients, and defense signals. Many studies suggest that MNs play a key role in **forest resilience**, **regeneration**, and **community stability**

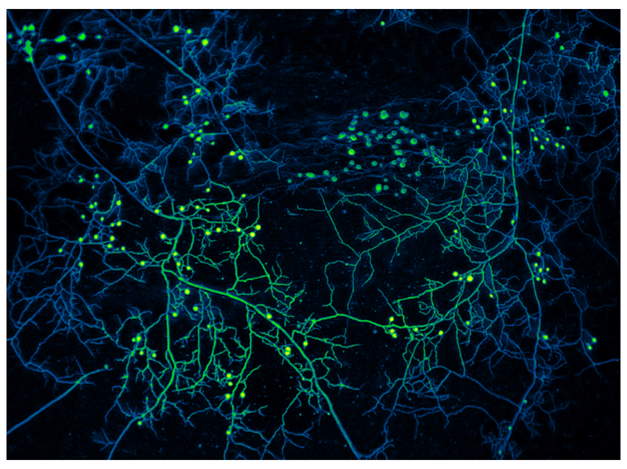


Diagram of Mycorrhizal fungus attached to plant root. [1]

### Why is the project important?

It still remains unclear how the structure of these networks influences feedback mechanisms and emergent ecosystem-level behavior.
This project explores these dynamics in a simplified modeling framework, with the aim of **providing conceptual insights relevant to ecosystem-related studies**.

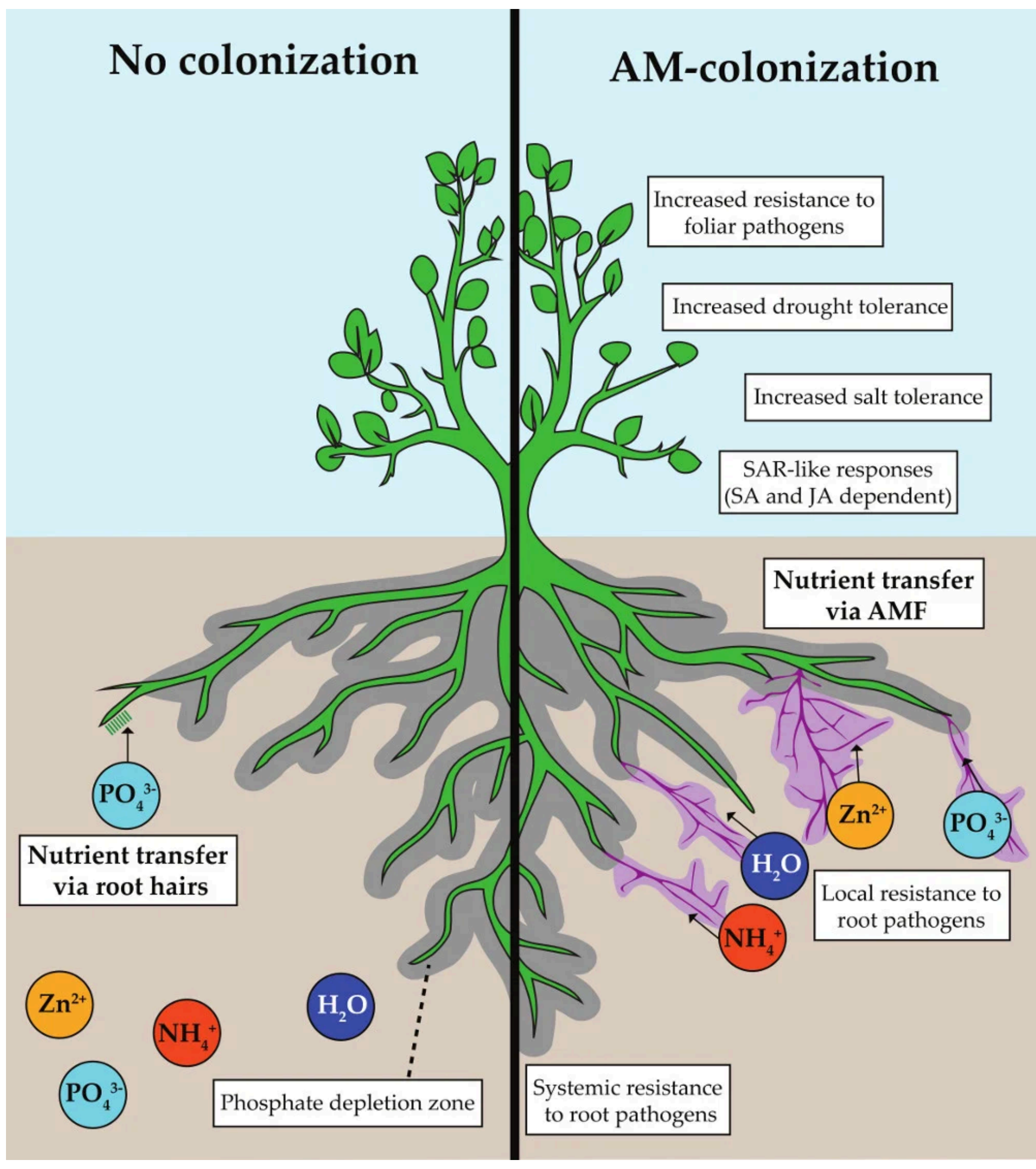


Diagram of fungal colonization. [2]

### Our Goal

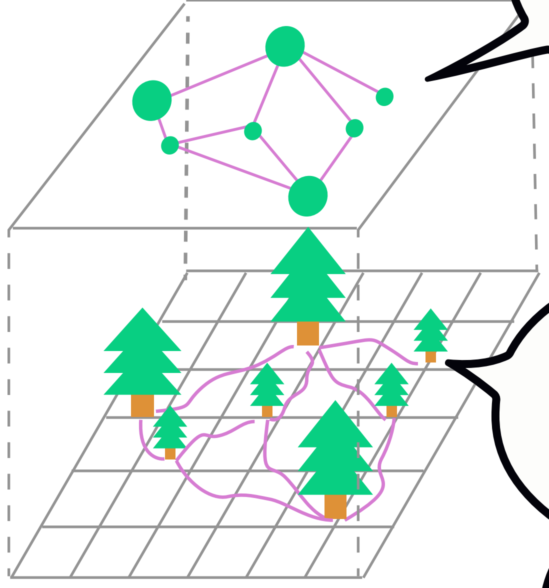
**Analyze Feedback Mechanisms:** Determine if network links stabilize forest communities and/or amplify change during stress.

**Evaluate Self-Organization:** Investigate how resource redistribution influences emergent growth patterns.

**Reproduce Experimental Results in a Simulation:** The simulation should produce results that conform to observed patterns and tendencies.

## 2. Model Structure

**Network Link - Fungus**
"Scale-free" network dynamics, with a power-law degree distribution. Carbon is allocated along these links.



**Agent-based Model**
Forest dynamics are simulated on an evolving square lattice. Tree growth, survival, and sapling establishment are governed by carbon uptake and environmental stress.

**Agents:**
- hubs
- seedlings

Our Theorized Model

**Preferential attachment** depending on the connectivity (degree), spatial distance, and biomass. We set the weight (likelihood) for connecting to a node as:

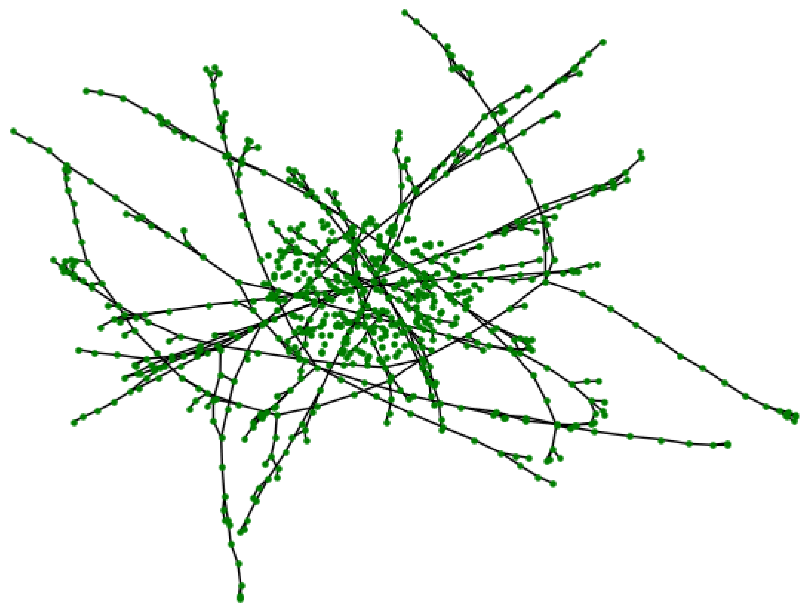
$$W_v = \underbrace{(k_v + \delta)^\beta}_{\text{topological advantage}} \times \underbrace{\exp\left(-\frac{\alpha \cdot d(u,v)}{B_v + \epsilon}\right)}_{\text{biophysical constraint}}$$

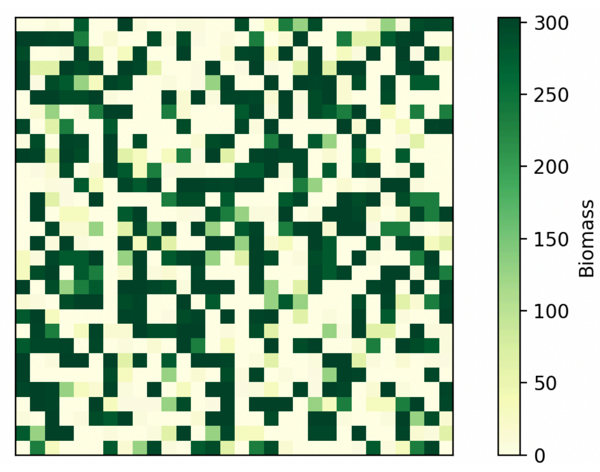
Carbon is redistributed within each connected component using a **capacity-constrained water-filling algorithm**.

The carbon acquisition of an individual at each step is calculated as:

$$C_{in}(i,j) = \frac{r \cdot B_{i,j}}{1 + E_{stress}}$$

Model dynamics

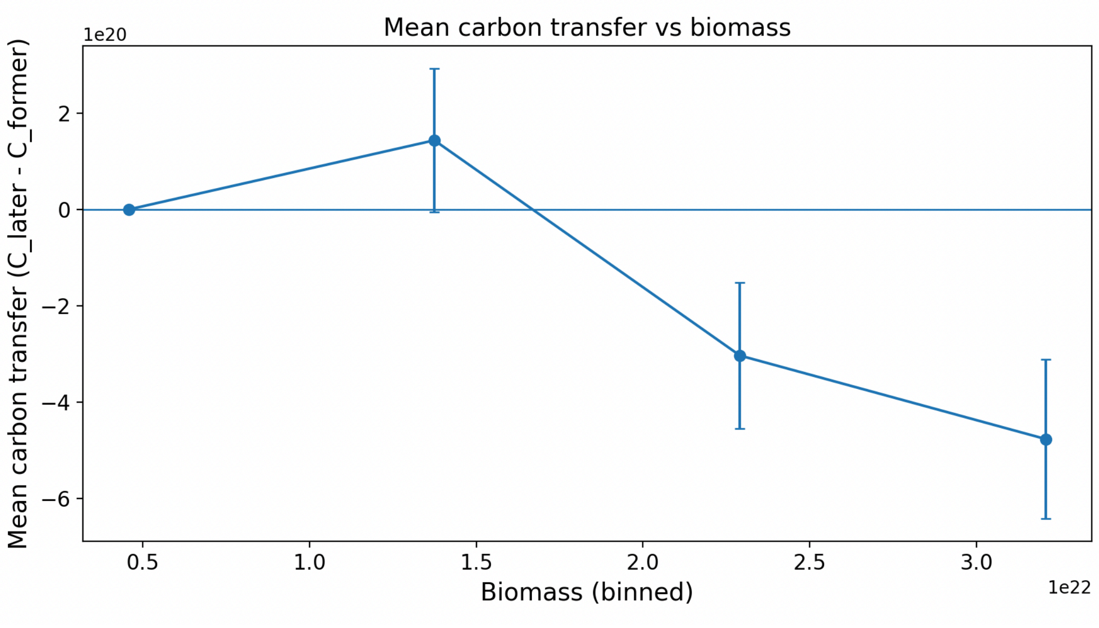


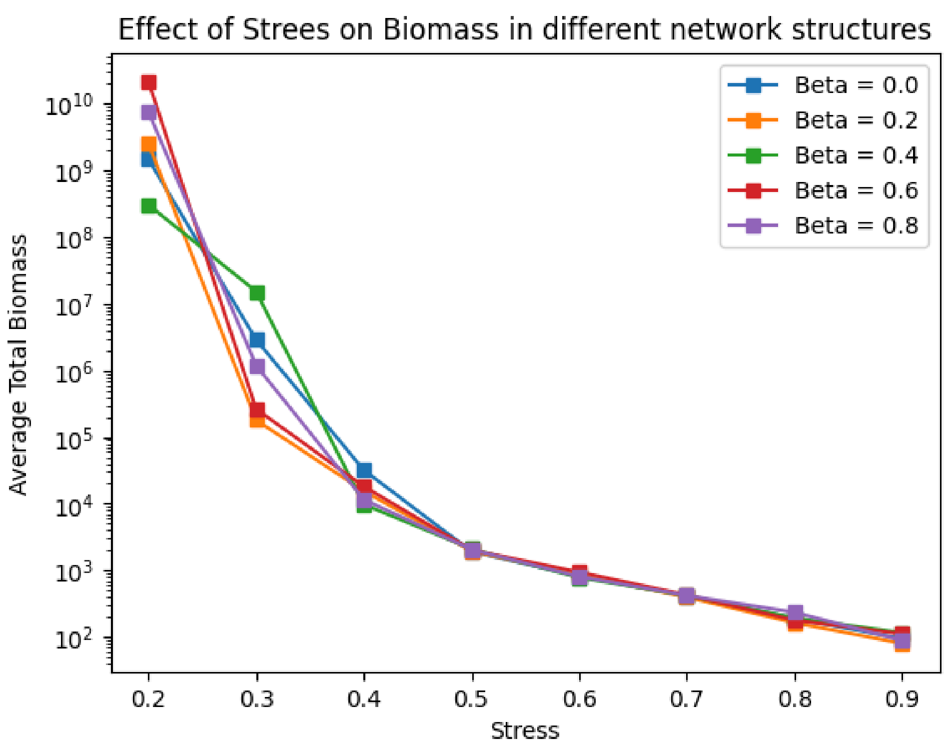


Plot of simulated structure

## 3. Results

The model successfully reproduces the **"source-sink" dynamics** of MNs, where nutrients are redistributed across the network. New seedlings connect to the MNs and draw resources from the more established "hub trees." Under normal conditions, the nutrient benefit received from the MN decreases as the tree ages.




These dynamics change depending on MN structure, nutrient density and environmental stressors. Additionally, the presence of diverse MNs greatly increases forest resiliency, allowing new growth to repopulate damaged areas fast.

### Take-home Message

- The nutrient benefit received from the MN decreases as the tree ages, helping young trees to grow.
- Harsh environments make it much harder for trees to survive, but different MN structures can result in different properties.

**References**
[1] Oyarte-Galvez (AMOLF), Mycelium of arbuscular mycorrhizal fungi with false color,,CC BY-SA 4.0
[2] Catherine N.Jacob C, Jeremy D. Murray and Christopher J. Ridout, CC BY-SA 4.0, via Wikimedia Commons